

Original Contribution

Transcutaneous electrical acupoint stimulation before surgery reduces chronic pain after mastectomy: A randomized clinical trial

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ABSTRACT

Study objective: Despite multiple interventions, the incidence of chronic pain after mastectomy could be as high as 50% after surgery. This study aimed to determine the efficacy of transcutaneous electrical acupoint stimulation (TEAS) before anesthesia induction in reducing chronic pain and to compare the effect of combined acupoint TEAS with that of single acupoint TEAS.

Design: A multicenter randomized clinical trial.

Setting: The study was conducted at six medical centers in China from May 2016 to April 2018. Final follow-up was on October 26, 2018.

Participants: Eligible patients were women scheduled for radical mastectomy under general anesthesia.

Interventions: Patients were randomly and equally grouped into sham control ($n = 188$), single acupoint (PC6, $n = 198$), or combined acupoints (PC6 and CV17, $n = 190$) TEAS groups using a centralized computer-generated randomization system. TEAS was applied for 30 min before anesthesia induction. The sham-operated control group received electrode attachment but without stimulation. Anesthesiologists, surgeons, and outcome assessors were blinded to the interventions.

Measures: The primary endpoint was the incidence of chronic pain 6 months after surgery. Incidences were compared among the groups using the unadjusted χ^2 test.

Results: Of the 576 randomized patients, 568 completed the trial. In the intention-to-treat analysis, post-mastectomy pain at 6 months was reported in 42 of 190 patients (22.1%) in the combined acupoints group, 65 of 188 patients (34.6%) in the sham-operated group ($P = 0.007$; relative risk [RR], 95% confidence interval [CI]: 0.68, 0.52–0.89), and 72 of 198 patients (36.4%) in the single acupoint group ($P = 0.002$; RR, 95% CI: 0.72,

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0.55–0.93). Remifentanyl consumption during surgery and postoperative nausea and vomiting at 24 h after surgery were lower in the combined acupoint group than that in the sham-operated group.

Conclusion: TEAS at combined acupoints before surgery was associated with reduced chronic pain 6 months after surgery.

Trial registration: [Clinicaltrials.gov](https://clinicaltrials.gov) identifier: NCT02741726. Registered on April 13, 2016.

1. Introduction

Persistent postsurgical pain is clinically relevant in patients undergoing mastectomy, which causes both physical discomfort and daily dysfunction. The compromised daily function of the patients affects the entire society to operate effectively and economically [1]. In a recent study by Sessler et al. [2], incision pain at 6 months after mastectomy and chronic persistent neuropathic pain was reported in 52% and 10% of the patients, respectively. At least 50% of patients had mild pain, and 16% had moderate to severe pain, 1 year after lumpectomy and mastectomy [3].

Current treatments for persistent pain after mastectomy include gabapentinoids, antidepressants, regional nerve block, laser therapy, and thermal radiofrequency of the stellate ganglion [4]; however, no cure has been guaranteed. Previous studies demonstrated that acute pain and anesthesia management is closely associated with the development of chronic pain [5,6]. We hypothesized that adequate management of acute pain and anesthesia might be a suitable target to reduce chronic pain after mastectomy. Acupuncture and related techniques such as transcutaneous electrical acupoint stimulation (TEAS) have been shown to reduce the consumption of opioids during general anesthesia [7,8] and are used to treat acute post-surgical pain [9,10]. Therefore, this multicenter, prospective, randomized, controlled trial aimed to investigate the effect of TEAS before anesthesia induction on post-mastectomy chronic pain and to compare the efficacy of combined acupoint TEAS with that of single acupoint TEAS.

2. Materials and methods

2.1. Study overview

The study was conducted at six medical centers and was approved by the institutional review boards of Xijing Hospital, Fourth Military Medical University; First Affiliated Hospital, Xi'an Jiaotong University; Tangdu Hospital, Fourth Military Medical University; First Affiliated Hospital, Zhengzhou University; First Affiliated Hospital, Chongqing Medical University; and People's Hospital of Zhengzhou University. It was registered at [ClinicalTrials.gov](https://clinicaltrials.gov) (NCT02741726). Written informed consent was obtained from each participant before the study. The protocol adherence of the six participating centers was assessed by an independent observer through regular visits. The detailed protocol of this clinical trial is also submitted as a supplemental file.

2.2. Patient characteristics

Women aged 18–65 years who were scheduled for elective radical mastectomy under general anesthesia were recruited for this study. Exclusion criteria included contraindications to TEAS (scar or skin damage at the stimulation site, electronic devices such as pacemakers or other implanted medical electronic devices in the body); communication difficulties; history of breast surgery; drug abuse/addiction; or alcohol abuse or addiction; cardiac dysfunction or severe hypertension; confirmed liver/renal dysfunction; giving consent for another interventional study; or declined participation.

2.3. Randomization and masking

Study participants were randomized 1:1:1 into the sham-operated,

single acupoint (TEAS at PC6), and combined acupoint groups (TEAS at PC6 and CV17) using a secured web-based system that was stratified according to permuted blocks (random block sizes; maximum of 12) by the treatment center. Outcomes were assessed by trained investigators who were blinded to the treatment allocation. The patients and investigators who participated in the intervention were not masked. Other clinical staff, including surgeons, anesthesiologists, and ward staff, were unaware of the allocation.

2.4. Study intervention

Transcutaneous acupoint stimulation was performed in compliance with recommendations from the Standards for Reporting of Controlled Trials in Acupuncture [11]. Briefly, electrodes were attached to the skin surface of the acupoint and connected to the Hwato Electric Acupuncture Treatment Instrument by leads (model No. SDZ-V; Suzhou Medical Appliances Co., Ltd., Suzhou, China; see Supplemental Material 1). Patients in the combined acupoints group received TEAS at bilateral PC6 (Neiguan, a key acupoint of the hand-jueyin pericardium meridian) and CV17 (Danzhong, a key acupoint of the Ren meridian). Patients in the single acupoint group received TEAS at the bilateral PC6 acupoints (see Supplemental Material 1). Patients in the sham-operated group received electrode attachment but without stimulation. The device provided “disperse-dense” waves with alternating frequencies of 2 Hz and 10 Hz for two cycles. The stimulation intensity was identified as the maximal tolerance to the “Teh Chi” sensations of heaviness, numbness, and swelling at the point of stimulation. Anesthesia induction was initiated after 30 min of TEAS.

Interventions were performed by a designated investigator who was not involved in the anesthesia or follow-up. The stimulator was placed in an opaque box to blind the surgical team and anesthesiologist for group allocation.

2.5. Patient-reported outcome measures

At 3 months and 6 months after surgery, the patients were followed up using a telephonic questionnaire [6]. The primary endpoint was the incidence of post-mastectomy chronic pain at 6 months after surgery, which was defined as chronic pain in the breast scar area, axilla, arm, and side of the body. The questionnaire is provided in Supplemental Material 2.

The secondary endpoints included remifentanyl consumption during general anesthesia, the time to the first verbal response and the time to endotracheal extubation, postoperative nausea and vomiting (PONV), respiratory depression, numeric rating scale (NRS) scores, and demand for rescue analgesics, patient satisfaction scores on analgesia by 24 h after surgery, and the incidence of chronic pain at 3 months after surgery. In case of pain at 3 months or 6 months after surgery, the severity of chronic pain was assessed using the NRS [6,12].

2.6. Procedures

All surgical procedures were performed under total intravenous anesthesia with endotracheal intubation. Anesthesia was induced with midazolam, propofol, and fentanyl, and continuous infusion of remifentanyl and propofol was titrated to maintain a Narcotrend index of 40–60 and stable hemodynamics. After endotracheal extubation, the patients were transferred to the post-anesthesia care unit. Forty

milligrams of parecoxib sodium was administered before the incision. Sufentanil was used for patient-controlled analgesia after surgery. Parecoxib sodium (40 mg) was administered as rescue analgesia for those whose NRS was higher than 4. At 3 months and 6 months after surgery, the patients were followed up using a telephonic questionnaire to assess chronic pain [6]. The patient was inquired regarding pain. If the patient answered yes, the pain intensity, neuropathic components (modified Douleur Neuropathique 4 [DN4] questionnaire) [12], and quality of recovery were assessed. Investigators who completed the telephonic follow-up were blinded to group allocation.

2.7. Statistical analysis

The primary hypothesis was that TEAS would reduce chronic pain at 6 months after mastectomy, and combined acupoint stimulation would be more effective than single acupoint stimulation. We calculated that a sample size of 537 participants would have 80% power to detect a difference in the incidence of chronic pain at 6 months after mastectomy among the combined acupoints (25%), single acupoint group (30%), and sham-operated groups (40%), using two degrees of freedom chi-square test with a significance alpha level of 0.05. Considering a dropout rate of 5%, we defined a sample size of 576 participants.

We performed all the analyses in an intention-to-treat population, which included all patients who had undergone randomization and interventions. Descriptive statistics were applied to present the characteristics of each group with mean $\pm$ standard deviation, median (interquartile), or the number of cases (%). Either analysis of variance or Chi-square test was used to reveal the baseline balance between groups, depending on the characteristics of the variables. The incidence of post-mastectomy chronic pain at 3 months and 6 months after surgery was

compared with an unadjusted chi-square test. Logistic regression models with log links to estimate relative risk (RR) were used to assess RR with a 95% confidence interval (CI). Multiple comparisons between two groups were performed using the Chi-square partition method, in which the type I error was corrected to 0.0125 by the Bonferroni-adjusted method. Statistical significance was set at a two-tailed *P*-value of 0.05. In a post-hoc analysis, we assessed the heterogeneity of the treatment effect on post-mastectomy chronic pain at 6 months after surgery across study sites by testing for the treatment-by-site interaction. An interaction term <0.10 suggested potential treatment effect heterogeneity across sites. For missing data, analysis was performed according to a worst-case scenario. In particular, patients lost to follow-up in the two TEAS groups and those in the sham group were considered with and without pain, respectively. Stratified analysis by age, pre-existing pain at the surgery site, chemotherapy, radiotherapy, and endocrine therapy were performed using the Cochran-Mantel-Haensel test. All statistical analyses were conducted using Statistical Analysis System (SAS) version 9.2 (SAS Institute Inc., 2010).

3. Results

3.1. Patient characteristics

From May 2016 to April 2018, 1593 patients were screened in six medical centers (Fig. 1). A total of 576 patients met the criteria and were randomly assigned into the sham-operated control ($n = 188$), single acupoint ($n = 198$), and combined acupoints ($n = 190$) groups (Fig. 1). Four patients discontinued the intervention owing to a change in the surgery. Another four patients were lost to follow-up. Therefore, 568 patients were included in the study. No significant difference was found

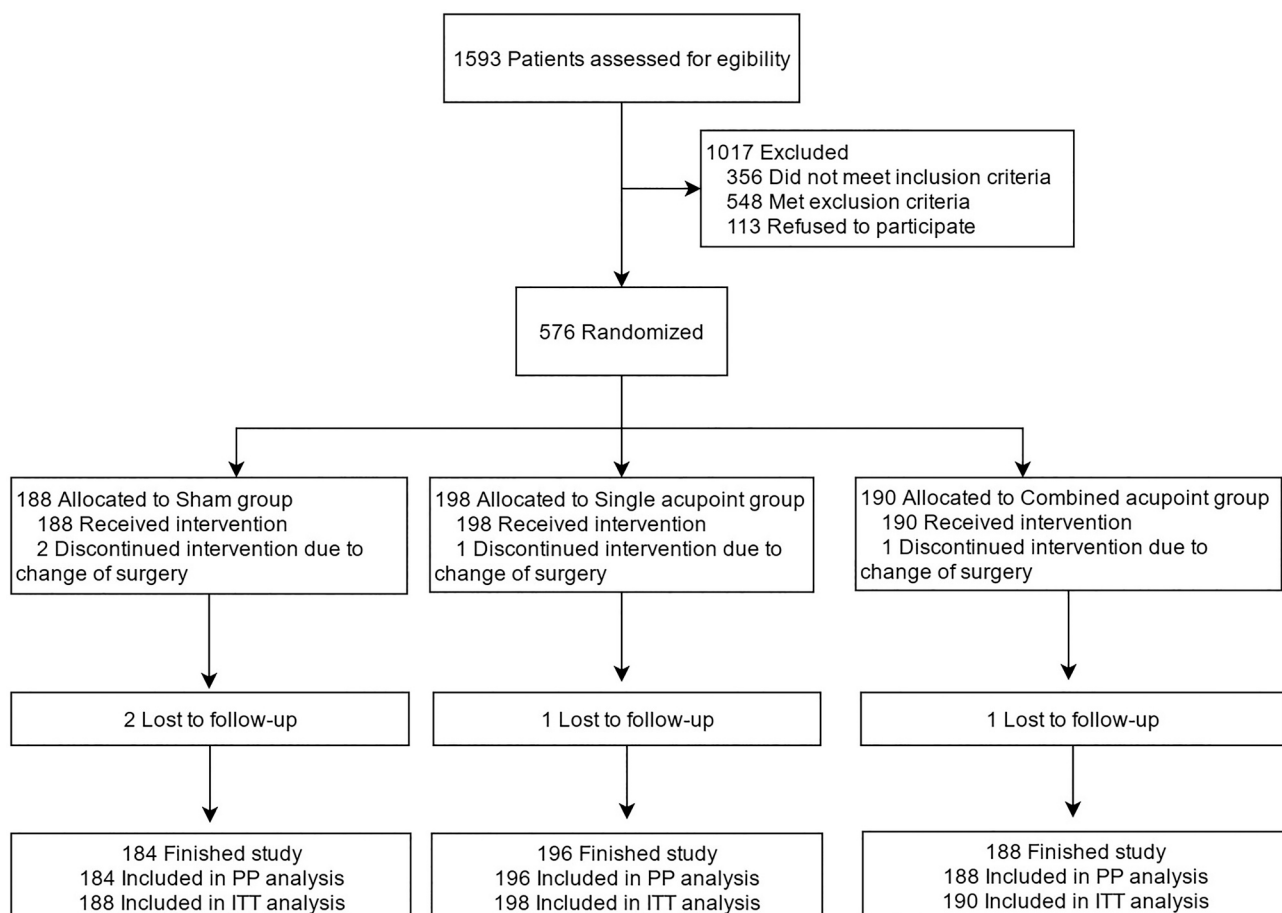


Fig. 1. Flow chart of participants in the trial.

in the dropout rate between the groups ($P = 0.8$). Patient baseline characteristics, anesthetic and surgical parameters, and postoperative treatments did not differ among the groups (Table 1). Patient characteristics were well balanced by the study site and clinical factors.

3.2. Primary outcome

In the intention-to-treat analysis, post-mastectomy chronic pain at 6 months after surgery was reported in 42 of 190 patients (22.1%) in the combined acupoints group, which was significantly lower than that in 65 of 188 patients (34.6%) in the sham-operated group ($P = 0.001$; RR, 95% CI: 0.68, 0.52–0.89) and 79 of 196 patients (40.3%) in the single acupoint group ($P = 0.001$; RR, 95% CI: 0.72, 0.55–0.93). The treatment effects were homogeneous across the trial sites. Analysis of the primary outcome stratified by age, pre-existing pain at the surgery site, chemotherapy, radiotherapy, and endocrine therapy showed no influence of these factors (Supplemental Material 4). However, the NRS score of pain provided by patients who reported pain at 6 months after surgery did not differ among the groups ($P = 0.31$) (Table 2).

3.3. Secondary outcomes

Remifentanyl consumption during surgery was significantly lower in the combined acupoints group than in the sham group (Table 3, $P < 0.001$, group difference 95% CI: $-0.21 [-0.34, -0.07]$); however, it was not different between the single acupoint and sham groups ($P = 0.04$). The time to first verbal response and endotracheal extubation in the combined acupoints and single acupoint groups did not differ compared to the sham control group ($P = 0.80$ and 0.82 , respectively). The

incidence of PONV at 24 h after surgery was 18.4% in the combined acupoints group, which was significantly lower than that in the single acupoint group (29.8%, $P = 0.004$; RR, 95% CI: 0.71 [0.53, 0.94]) and sham group (36.2%, $P < 0.001$; RR, 95% CI: 0.60 [0.45, 0.81]). The severity score of PONV in the combined acupoints group was lower than that in both the single acupoint and sham groups ($P < 0.001$). However, further analysis revealed that the incidence of nausea differed among the groups ($P < 0.001$), but not the incidence of vomiting ($P = 0.93$). The NRS scores at rest and cough did not differ among the three groups ($P = 0.12$ and 0.49). The demand for rescue analgesics and patient satisfaction score for analgesia were similar among the groups ($P = 0.58$) (Table 3).

Post-mastectomy pain at 3 months after surgery was reported in 58 of 190 patients (30.5%) in the combined acupoints group, 75 of 188 patients (39.9%) in the sham group, and 70 of 198 patients (35.4%) in the single acupoint group (Table 2). There were no significant differences among the three groups. However, for patients with chronic pain at 3 months post-surgery, the NRS scores showed a significant difference ($P = 0.04$). Pairwise comparison showed that patients in the combined acupoints group had a lower NRS score at 3 months after surgery than that in the sham-operated group ($P = 0.03$, mean difference, 95% CI: $-0.56 [-1.01, 0.11]$). At 6 months after surgery, patients with chronic pain in the three groups had similar NRS scores. Moreover, there was no difference in the number of patients with the neuropathic component, disability of the ipsilateral limb or shoulder, the problem of self-care and usual activities, or seeking medical management of pain among the three groups both at 3 months and 6 months after surgery. The results of the per-protocol analysis did not differ from those of the intention-to-treat analyses (Supplemental Material 3).

3.4. Adverse events

One patient in the single acupoint group reported discomfort in the skin area attached to the electrodes. No differences were found among the three groups.

4. Discussion

This multicenter, prospective, randomized, controlled trial revealed that TEAS at combined acupoints before anesthesia induction in patients undergoing mastectomy is associated with a decreased incidence of chronic pain at 6 months after surgery compared with the sham intervention. This discovery may have potential clinical significance due to high prevalence of post-mastectomy chronic pain.

In our study, the incidence of chronic pain at 6 months after mastectomy was 34.6% in the sham-operated group. The incidence of post-mastectomy chronic pain varies in previous studies, depending on the population, surgical technique, and assessment methods. Postoperative chronic pain has been managed similar to acute pain, with inadequate analgesia for acute pain being a risk factor [13–16]. Furthermore, higher morphine consumption 24 h post-surgery was observed in patients with increased incidence and severity of chronic pain [6]. Due to the side effects of opioids, combined analgesia using non-opioids for acute pain management is a fascinating strategy. The observed efficacy of TEAS on post-mastectomy chronic pain highlighted its possible role in acute pain management before surgery for reducing chronic pain after surgery. The lower incidence of post-mastectomy chronic pain by pre-surgery TEAS observed in the present study could be related to the TEAS-induced opioid-sparing effect. Previous studies have reported that avoiding high-dose exposure to the short-acting opioid remifentanyl can be important for the prevention of postoperative chronic pain [16–18]. These results indicate that TEAS applied before surgery may be beneficial for the long-term efficacy of pain prevention. Acupuncture has also been used for the treatment of chemotherapy-induced peripheral neuropathy symptoms and joint pain related to aromatase inhibitors [10,19].

Table 1
Demographic and surgical characteristics.

	Sham group (n = 188)	Single acupoint group (n = 198)	Combined acupoint group (n = 190)	P value
Age (years)	48.2 ± 8.2	48.0 ± 9.1	48.2 ± 9.0	0.98
Body mass index (kg/m ²)	23.3 ± 2.8	23.7 ± 2.6	23.6 ± 2.5	0.28
ASA status (I/II)	135/49	144/52	135/53	0.28
Charlson score				0.29
0	180 (95.7%)	194(98.0%)	188(99.0%)	
1	5(2.7%)	3(1.5%)	2(1.1%)	
2	3(1.6%)	1(0.5%)	0(0.0%)	
Duration of surgery (min)	122.6 ± 48.3	124.2 ± 42.2	124.4 ± 45.3	0.92
Duration of anesthesia (min)	140.7 ± 49.2	141.3 ± 44.1	141.8 ± 46.3	0.97
Dose of propofol (mg)	693.7 ± 241.0	711.5 ± 251.9	707.8 ± 227.0	0.75
Fluid input during surgery (mL)				
Crystalloid input	1088.8 ± 409.7	1068.4 ± 363.0	1130.2 ± 394.8	0.29
Colloid input	159.4 ± 425.8	133.7 ± 231.2	146.8 ± 244.5	0.72
Blood loss (mL), median (min, max)	40.9 ± 17.2	32.7 ± 28.0	33.0 ± 23.9	0.83
Urine (mL), median (min, max)	355.7 ± 322.2	311.0 ± 290.6	385.2 ± 575.1	0.21
Preoperative pain of the surgical site	23(12.2%)	28(14.1%)	32(16.8%)	0.44
Postoperative treatment				
Radiation	78(41.5%)	75(37.9%)	79(41.6%)	0.70
Chemotherapy	103 (54.8%)	106(53.5%)	110(57.9%)	0.68
Endocrine therapy	68(36.2%)	64(32.3%)	73(38.4%)	0.45

Data were presented as mean ± SD, median (IQR) or number of patients (percentage).

Table 2

Postoperative long-term outcomes.

	Sham group (n = 188)	Single acupoint group (n = 198)	Combined acupoint group (n = 190)	P value among groups	P value		
					RR (95%CI) or mean difference (95% CI)		
					Dual vs sham	Dual vs single	Single vs sham
Chronic pain by 3 months after surgery							
Incidence	75(39.9%)	70(35.4%)	58(30.5%)	0.17	/	/	/
NRS	2.8 ± 2.0 (n = 75)	2.6 ± 2.0 (n = 70)	2.3 ± 2.0 (n = 58)	0.04	0.03 −0.56(−1.01, −0.11)	0.14 −0.28(−0.72, 0.17)	0.26 −0.29(−0.73, 0.16)
Neuropathic pain	15(8.0%)	19(9.6%)	19(10.0%)	0.77			
Disability of ipsilateral limb or shoulder	33(17.6%)	29(14.6%)	28(14.7%)	0.67			
Problem of self-care and usual activities	5(2.7%)	2(1.0%)	5(2.6%)	0.43			
Medical management of pain	5(2.7%)	2(1.0%)	3(1.6%)	0.45			
Chronic pain by 6 m after surgery							
Incidence	65(34.6%)	72(36.4%)	42(22.1%)	0.005	0.002 0.68(0.52,0.89)	0.007 0.72 (0.55,0.93)	0.71 0.95 (0.73,1.25)
NRS	2.8 ± 1.8 (n = 65)	2.7 ± 2.0 (n = 72)	2.4 ± 1.9 (n = 42)	0.31	/	/	/
Neuropathic pain	12(6.4%)	18(9.1%)	18(9.5%)	0.50			
Disability of ipsilateral limb or shoulder	16(8.5%)	18(9.1%)	22(11.6%)	0.56			
Problem of self-care and usual activities	1(0.5%)	0(0%)	2(1.1%)	0.37			
Medical management of pain	2(1.1%)	5(2.5%)	0(0%)	0.07			

Data were presented as mean ± SD, or number of patients (percentage). In multiple comparisons between each two groups, the type I error was corrected to 0.0125.

Table 3

Postoperative short-term outcomes.

	Sham group (n = 188)	Single acupoint group (n = 198)	Combined acupoint group (n = 190)	P value among groups	P value		
					Group difference (95%CI)		
					Dual vs sham	Dual vs single	Single vs sham
Remifentanyl consumption (mg)	1.4 ± 0.7	1.3 ± 0.6	1.2 ± 0.5	0.002	<0.001	0.09	0.04
					−0.20 (−0.34, −0.07)	−0.10 (−0.23,0.03)	−0.10 (−0.23,0.03)
Time to verbal response (min)	9.2 ± 5.1	9.1 ± 4.4	9.1 ± 4.9	0.80	/	/	/
Time to extubation (min)	10.7 ± 5.9	10.4 ± 4.8	10.6 ± 5.2	0.82	/	/	/
NRS at rest at 24 h after surgery	1(0, 1.6)	0.9(0, 1.3)	1(0,1)	0.94	/	/	/
NRS at cough at 24 h after surgery	1.1(0.1, 2.8)	1.3(0.3, 2.9)	1(0,2.3)	0.89	/	/	/
Rescue analgesia by 24 h after surgery	7(3.7%)	3(1.5%)	4(2.1%)	0.35	/	/	/
PONV by 24 h after surgery	68(36.2%)	59(29.8%)	35(18.4%)	0.0001	<0.001	0.009	0.20
					0.60(0.45,0.81)	0.71(0.53,0.94)	0.87(0.70,1.07)
Nausea by 24 h after surgery	45(23.9%)	38(19.2%)	12(6.3%)	<0.001	<0.001	<0.001	0.27
					0.38(0.23,0.63)	0.46(0.28,0.75)	0.87 (0.67,1.12)
Vomiting by 24 h after surgery	22(11.7%)	21(10.6%)	22(11.6%)	0.93			
Score of PONV	0.4 ± 0.7	0.4 ± 0.7	0.3 ± 0.7	0.03	<0.001 0.18 (0.02,0.34)	0.008 0.10 (−0.06,0.26)	0.19 0.08 (−0.08,0.24)
Patient's satisfaction score	8.2 ± 0.9	8.2 ± 1.1	8.4 ± 0.9	0.16			

Data were presented as mean ± SD, median (IQR) or number of patients (percentage). In multiple comparisons between each two groups, the type I error was corrected to 0.0125.

The effects of acupuncture on pain have been extensively investigated. Acupuncture may act through peripheral, spinal, and supraspinal mechanisms with the involvement of bioactive molecules, including opioids [20], 5-hydroxytryptamine [21], norepinephrine [22],

glutamate receptors and transporters [23], and signal molecules [24–27]. Of these, opioids have been proven to play a central role in electroacupuncture (EA) and TEAS for pain inhibition [27]. The release of opioid peptides evoked by EA is frequency-dependent. EA at 2 Hz and

100 Hz induces the release of enkephalin and dynorphin in the spinal cord, respectively [28]. Some studies on EA have shown that low-frequency stimulation is more effective against nociceptive pain and carrageenan-induced inflammatory pain [27,29,30]. In previous studies on acupuncture analgesia, both 2/10 Hz and 2/100 Hz protocols were frequently used [31]. In our previous studies, we used the 2/10 Hz protocol and found it to be effective for reducing opioid consumption in patients undergoing sinusotomy and ambulatory breast surgery. Therefore, in this study, we followed the stimulation protocol described in our previous studies. In the present study, we did not measure the levels of endogenous opioids. Further studies are needed to investigate the underlying mechanism of acupuncture before surgery to decrease postoperative chronic pain.

Several other risk factors, such as age, axillary surgery, radiotherapy, body mass index, preoperative pain at the surgical site, nerve damage, psychosocial factors, and genetic predisposition have been proposed to influence chronic pain after mastectomy [15]. In our study, baseline characteristics including age, surgical technique, pre-existing pain at the surgical site, and adjuvant therapies such as chemotherapy, radiotherapy, and endocrine therapy were not different among the groups.

Previous studies have shown that approximately 14% of patients experience long-term moderate-to-severe pain in the area of breast surgery [15]. In our study, we found a lower incidence of chronic pain at 6 months after mastectomy in the combined-acupoint TEAS group (22.1%) than that in the sham (34.6%) and single-acupoint TEAS groups (36.4%). We only compared the NRS scores of patients with chronic pain and found no differences among the groups. Although the NRS scores of patients with chronic pain at 3 months and 6 months after surgery were low (around 2) in all the groups, the pain-affected the movement of the ipsilateral limb or shoulder (Table 2).

There is no consensus on the number of acupoints that should be selected in acupuncture therapy. In real-world practice, acupuncturists decide the number of acupoints based on the theory of traditional Chinese medicine and their personal experiences. In the present study, we designed a protocol to compare the efficacy of TEAS between a single acupoint and combined acupoints to clarify whether TEAS at a single acupoint is sufficient to induce analgesic efficacy for long-term chronic pain after surgery. Unlike the efficacy of combined acupoint stimulation, although single acupoint TEAS could decrease remifentanyl consumption during surgery, it could not reduce the NRS score and incidence of chronic pain by 3 or 6 months after surgery. Our previous clinical trial found that TEAS with two acupoints showed a stronger sparing effect of sufentanil than that with a single acupoint during cardiac surgery under general anesthesia [32]. A limited number of clinical trials have suggested a better efficacy of acupuncture using acupoint combinations than using a single acupoint in the treatment of various diseases. In rats, Yan et al. found that EA at CV12 (Zhongwan), ST25 (Tianshu), and ST37 (Shangjuxu) were better at relieving acetic acid-induced intestinal mucosal lesions than EA at a single acupoint [33]. A study assessing PONV found that double-acupoint stimulation was better than single acupoint stimulation [34]. Furthermore, acupuncture on DU26 (Shuigou) and KI1 (Yongquan) accelerates the recovery of consciousness after general anesthesia, with a possible synergistic effect of DU26 and KI1 [35]. In the present study, P6 and RN17 were selected for combined acupoint stimulation. P6 is a frequently used acupoint for analgesia [30]. RN17 is the key acupoint of the Ren meridian. According to traditional Chinese medicine theory, RN17 is important for treating chest diseases, including those of the breast. Combined TEAS of P6 and RN17 significantly decreased pain and PONV compared to the sham intervention and single acupoint stimulation, indicating the necessity of using multiple acupoints in TEAS.

Imaging studies also provide evidence that suggest the use of combined acupoints [36,37]. By measuring functional brain activities, brain regions that responded to acupuncture at different acupoints were identified. Using functional resting-state magnetic resonance imaging, Zhang et al. observed an increased number of brain regions with changes

in their activities and stronger activation after performing acupuncture at the LR3 (Taichong) and KI3 (Taixi) acupoints compared with performing acupuncture at LR3 alone [38]. An observational study found that general anesthesia in combination with triple-acupoint TEAS during radical mastectomy significantly decreased the demand for analgesics and improved acute pain relief. An increased level of plasma beta-endorphin was observed, which might be one of the underlying mechanisms [39].

4.1. Limitations of the study

This study had some limitations. First, TEAS before anesthesia induction could not be administered in a patient-blinded manner. The lack of blinding may have introduced bias to some extent in outcome monitoring. However, the investigators who performed the outcome assessment were blinded to group allocation. The stimulator was placed in an opaque box so that the surgeons and anesthesiologists were blinded. Electrodes were attached, and the threshold for stimulation was tested for all patients. Each patient was informed regarding the possible stimulation during TEAS. These procedures may minimize bias. Second, we excluded patients older than 65 years, which might have affected the generalizability of our results. Age has been significantly associated with chronic pain after breast surgery; young age is associated with higher risk. Patients aged 18–39 years and those aged 40–49 years have the highest risk of pain after breast conservative surgery and mastectomy, respectively [14]. Thus, our results indicate that patients with a higher risk of chronic pain may benefit from acupoint stimulation. Third, we did not compare the effects of acupoint stimulation with that of non-acupoint stimulation, owing to our previous study findings in which we verified that non-acupoint stimulation could not induce a remifentanyl-sparing effect in patients undergoing sinusotomy [7]. The specificity of acupoints has been proven in another previous study [40].

5. Conclusion

In patients undergoing radical mastectomy under general anesthesia, TEAS at combined acupoints before anesthesia induction is associated with less chronic pain 6 months after surgery than sham intervention.

Details of authors' contributions

Lize Xiong had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Acquisition, analysis, or interpretation of data: Jianfeng Jiang, Yonghui Wang, Lini Wang, Yaomin Zhu, Lanlan Zheng, Yuanyuan Guo, Liyuan Zhang, Tingting Liu, Xue Yang, Nian Zhang, Mingyang Sun, Junbao Zhang, Weixian Zhao.

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Disclosure

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Declaration of Competing Interest

All authors have completed the Unified Competing Interest form and declare: no support from any organisation for the submitted work; no financial relationships with any organisations that might have an interest in the submitted work in the previous three years, no other relationships or activities that could appear to have influenced the submitted work.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclinane.2021.110453>.

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